The associations of diet with serum insulin-like growth factor I and its main binding proteins in 292 women meat-eaters, vegetarians, and vegans.

✣ ✣ ✣

The lower rates of some cancers in Asian countries than in Western countries may be partly because of diet, although the mechanisms are unknown. The aim of this cross-sectional study was to determine whether a plant-based (vegan) diet is associated with a lower circulating level of insulin-like growth factor I (IGF-I) compared with a meat-eating or lacto-ovo-vegetarian diet among 292 British women, ages 20-70 years. The mean serum IGF-I concentration was 13% lower in 92 vegan women compared with 99 meat-eaters and 101 vegetarians ($P = 0.0006$). The mean concentrations of both serum IGF-binding protein (IGFBP)-1 and IGFBP-2 were 20-40% higher in vegan women compared with meat-eaters and vegetarians ($P = 0.005$ and $P = 0.0008$ for IGFBP-1 and IGFBP-2, respectively). There were no significant differences in IGFBP-3, C-peptide, or sex hormone-binding globulin concentrations between the diet groups. Intake of protein rich in essential amino acids was positively associated with serum IGF-I (Pearson partial correlation coefficient; $r = 0.27$; $P < 0.0001$) and explained most of the differences in IGF-I concentration between the diet groups. These data suggest that a plant-based diet is associated with lower circulating levels of total IGF-I and higher levels of IGFBP-1 and IGFBP-2.